AnimLab AI instructions – Gemini test report

Nov 3, 2026 · @Balázs

Summary

With the first instructions (v1), Gemini 3.5 Flash-Lite passed 3 of 6 tests; with the corrected instructions (v2) it passed all 5 re-tested prompts, including a new mechanics task. In v1, test 3 (projectile) failed outright, tests 1 and 2 got the physics right but drew badly, and test 6 needed a reworded prompt. Every fault traced back to a rule that v1 stated weakly or not at all. v2 adds those as numbered rules, a code pattern for moving arrows, a recipe for measuring with the physics world and a fourth tested example. What remains is drawing detail: wires that stop near, not exactly at, a component, and one Hungarian-only label.

Method

Each Gemini code and each reference code was run in AnimLab with the same slider values, and the readouts were compared with values calculated by hand.

Item	Detail
Model	Gemini 3.5 Flash-Lite, signed out, free tier at gemini.google.com
Session	One continuous chat, 2026-11-03 21:53–22:09 (Budapest), driven by Claude in the Claude desktop app's built-in browser
Instructions	`animlab-ai-instructions.md` v1, pasted as the first message (22,524 characters, checksum checked against the file)
Prompts	The six prompts of the test scenario, sent unchanged; test 5 and test 6 needed extra turns (see Findings)
Reference	Claude wrote its own code for every prompt with the same instructions, without looking at Gemini's answers first

Item	Detail
Run and check	A script loaded each code into AnimLab, set the sliders, waited, read the readouts and took a screenshot; a second script checked each code against the instruction rules

The Gemini codes in the project folder are byte-identical to what Gemini returned (length and checksum checked).

Results

Gemini's physics was right in four of six tests; test 3 gave a wrong measured range, and tests 1 and 2 had drawing faults. The reference codes matched the hand values in every test.

Test	Slider values	Expected	Gemini	Reference	Gemini verdict
T1 Ohm's law	U = 12 V, R = 6 Ω	I = U/(R + R_L), P = I²·R_L	I = 0.57 A, P = 4.90 W (R_L = 15 Ω)	I = 0.667 A, P = 5.33 W (R_L = 12 Ω)	Partial: physics right, drawing faults
T2 Transformer	U1 = 230 V, N1 = 500, N2 = 50, R = 10 Ω	U2 = 23 V, I2 = 2.3 A, I1 = 0.23 A	23.0 V, 2.30 A, 0.23 A	23.00 V, 2.300 A, 0.230 A	Partial: values right, graph and arrows wrong
T3 Projectile	v0 = 10 m/s, 45°	H = 2.55 m, D = 10.19 m	H = 2.91 m, D = 0.02 m	H = 2.54 m, D = 10.18 m	Failed
T3 Projectile	v0 = 20 m/s, 30°	H = 5.10 m, D = 35.31 m	H = 5.45 m, D = 0.06 m	H = 5.08 m, D = 35.28 m	Failed
T4 No-load switch	as T2, switch on	I2 = 0, I1 = 0 (ideal)	0.00 A, 0.00 A	0.000 A, 0.000 A	Passed
T5 Error fix	missing `)` on line 55	error found, full code back	found and fixed after it saw the code	same fix	Passed
T6 Rotor photo	n = 60 1/min	ω = 6.283 rad/s, picture turns about its centre	6.28 rad/s, turns correctly	6.283 rad/s, turns correctly	Passed after rewording

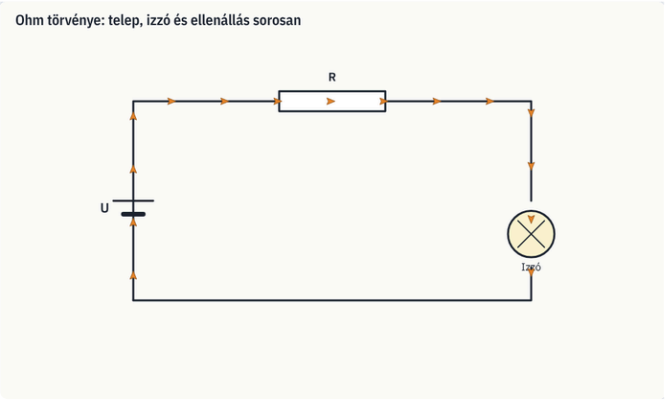
The two T1 currents differ only because the bulb resistance is a free choice (Gemini 15 Ω, reference 12 Ω); both follow I = U/(R + R_L). The scenario's own check values (I = 2 A, P = 24 W) assumed a bulb with no resistance, which was a mistake in the scenario.

Findings per test

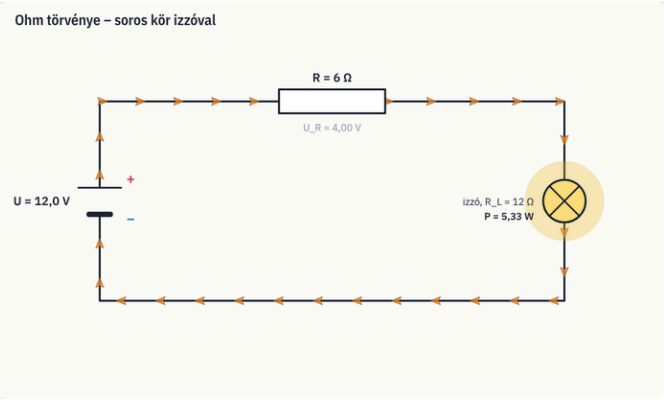
In every picture Gemini's scene is on the left and the reference on the right, both with the same slider values.

T1 – Ohm's law

T1 - Gemini 3.5 Flash-Lite

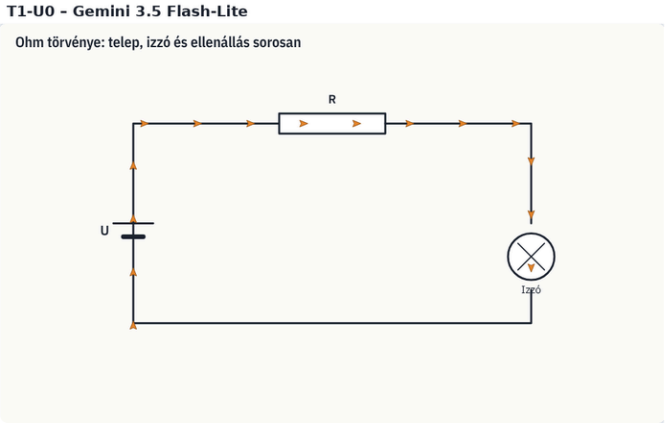


T1 - reference (Claude)

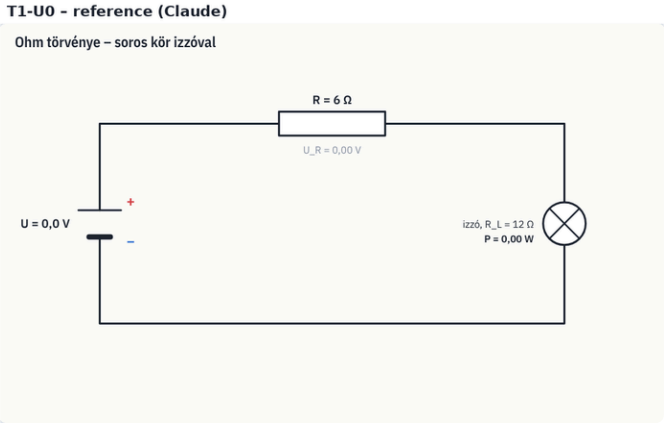


- **Right:** the bulb has its own resistance (15 Ω) and I = U/(R + R_bulb), P = I²·R_bulb; the earlier run in the scenario had missed this.
- **Wrong:** the wire runs straight through the battery plates; one arrow path runs across the resistor and the bulb; the bottom wire has no arrows; there is a gap between the bulb and the bottom wire.
- **Wrong:** the arrow speed changes with the slider but is not integrated with `pos`, so the arrows jump when U or R moves.

T1-U0 - Gemini 3.5 Flash-Lite

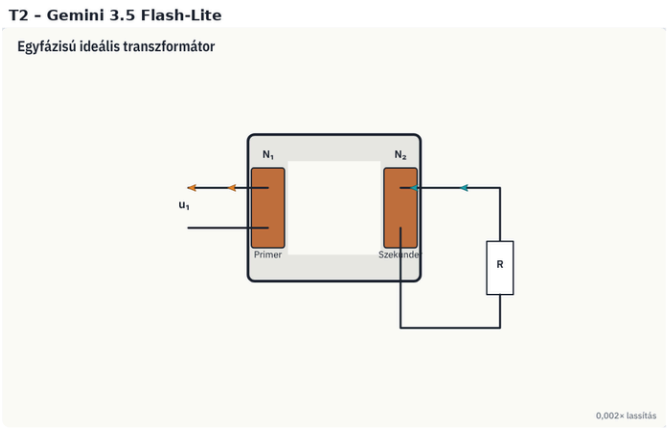


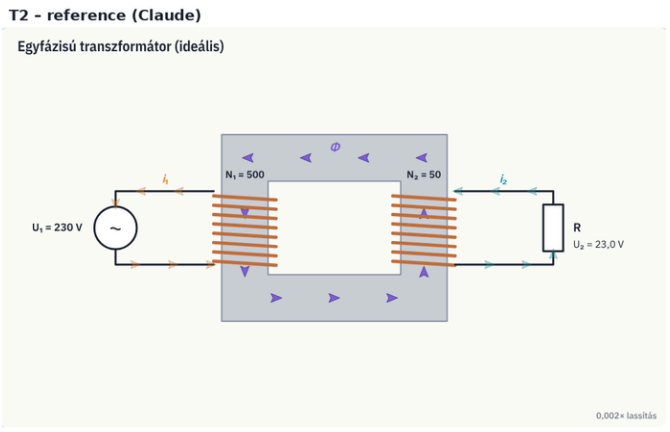
T1-U0 - reference (Claude)



- **Wrong:** at U = 0 V the arrows stop but stay visible, which suggests a current that is not there. The reference hides them and fades them at small currents.

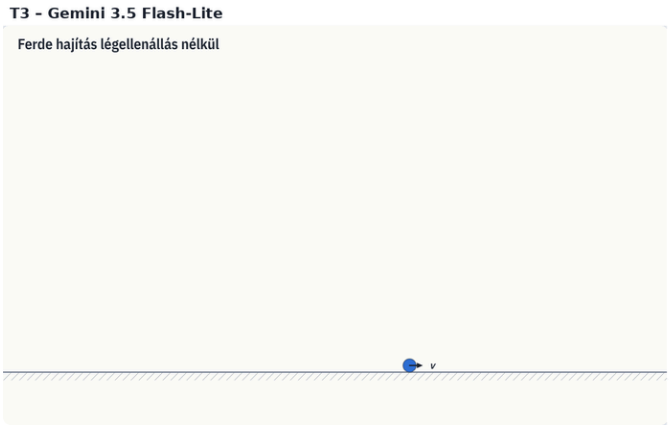
T2 – Transformer

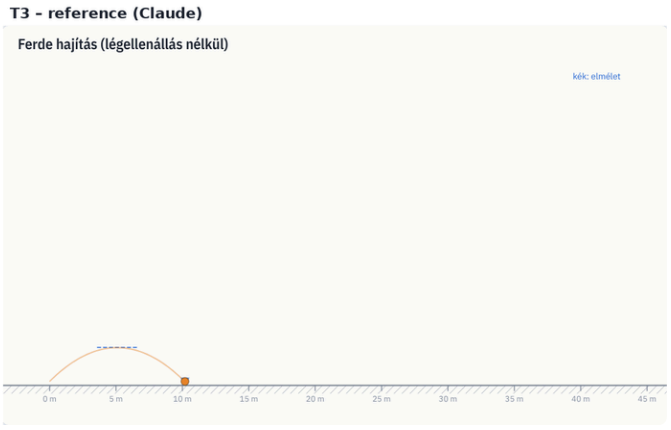




- **Right:** U2 = U1·N2/N1, I2 = U2/R, I1 = I2·N2/N1, and `timeScale(0.002)` for a 10 s period.
- **Wrong:** `plotWindow(0.4)` shows 20 periods, not the 2 asked for; it should be 2/f = 0.04 s.
- **Wrong:** the arrows use `speed: 80 * sin(wt)` without `pos`. With a varying speed the arrowhead offset is time × speed, so the arrows jump back and forth further each second.
- **Weak:** the wires end in the middle of the coils, the secondary return wire crosses the core, the coils are plain boxes, the flux is not shown, and only one wire of each circuit has arrows.

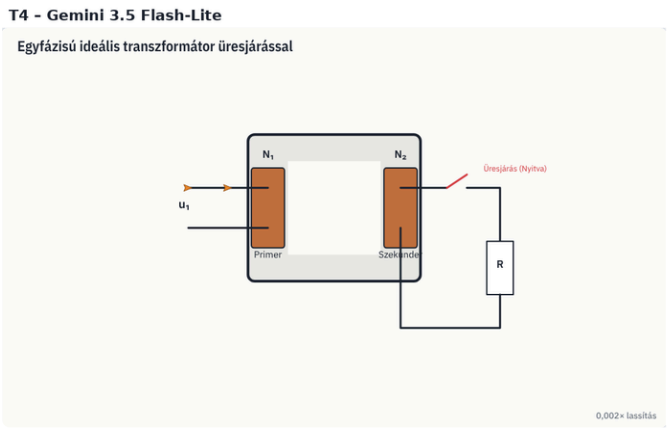
T3 – Projectile

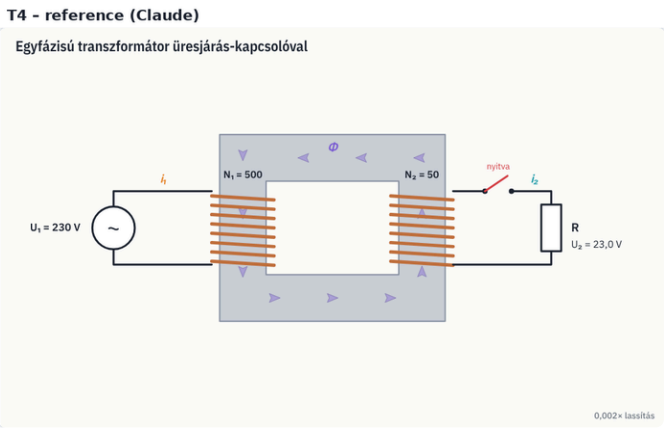




- **Wrong, the main failure:** the ball starts at y = 0, so its centre is on the ground and half of it underground. After the first step the world lifts it to y = r and marks it on the ground, the code reads that as landing, and the measured range freezes at 0.02 m.
- **Wrong:** `trail: true` instead of a number of points, so no trail is drawn.
- **Wrong:** the measured height uses the ball's centre, so it is too high by the radius (2.91 m against 2.55 m).
- **Wrong:** with the default friction the ball rolls on after landing and leaves the screen.
- **Right:** `resetOn("v0", "angle")`, the theory formulas and the velocity vector.

T4 – No-load switch



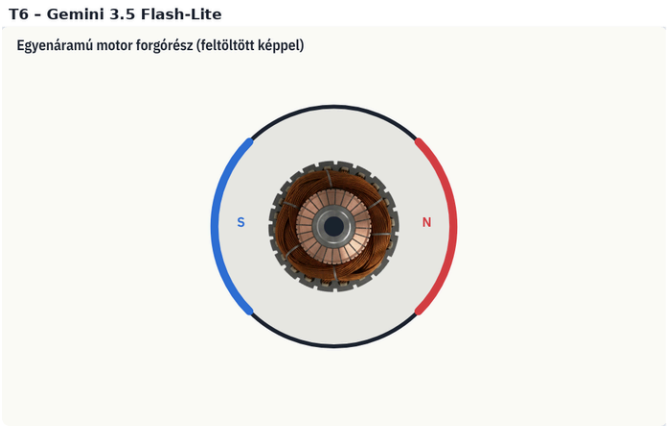


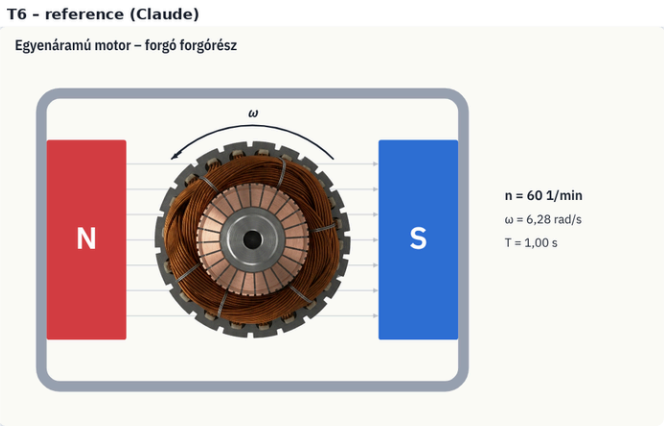
Gemini returned the whole script, added the toggle and drew an open switch, and I2 = I1 = 0. It kept slow primary arrows at no load, which matches a real transformer's magnetising current but not the ideal model it computes. The T2 faults (plot window, arrows without `pos`) stayed.

T5 – Error fix

Given only the error message, Gemini asked for the code, which is right because the error was planted by hand. Given the code, it named line 55 and the missing `)`, and returned the full script with that fix and nothing else changed except a typo in a comment.

T6 – Rotor photo





The first wording ("…a forgórész egy feltöltött kép legyen…") was taken as a request to generate an image, and Gemini answered that it cannot create images. After "Nem képet kérek, hanem AnimLab-jelenetet" it wrote a correct scene: it told the user to upload the picture as "rotor", integrated the angle from n, and the picture turns about its centre. Its N and S are only arcs on the housing and there is no direction arrow, but nothing is wrong.

Root causes and corrections

Most faults trace back to rules the v1 instructions stated weakly, in passing, or not at all, so v2 states each as a numbered rule and a checklist item.

Fault seen	Tests	Cause in v1	Correction in v2
Arrows jump: varying `speed` without `pos`	T1, T2, T4	The rule was one sentence in the `flow` table row, and example 5.1 itself used a varying `speed`	A code pattern in section 1 (integrate `pos` in `update`, `dir: sign(I)`); example 5.1 rewritten to use `pos`; checklist item
Ball starts underground, landing detected at once, measured range 0.02 m	T3	Section 3.5 did not say that `x, y` is the ball's centre or how to detect landing	`ball` row: "a ball on the ground starts at `y: r`"; a measuring recipe (height = `b.y - b.r`, landing after some flight time, `b.fixed = true`); new worked example 5.4
`trail: true` draws no trail	T3	The option was listed without a type	"`trail` is a number of points"
Graph shows 20 periods instead of 2	T2, T4	No rule linked `plotWindow` to frequency	"N periods: `plotWindow(N / f)`" in section 2 and the checklist
Wires through the battery and coils, arrows across components	T1, T2	No rule	Rule 11: wires stop at components, draw order wires → arrows → components
Arrows visible at zero current	T1	No rule	Rule 12: zero means nothing to show; fade in small values
Lamp glow invisible at default values	T1 (first run)	No rule	Rule 13: effects visible at defaults, scaled to the default case
Lamp resistance ignored	T1 (first run)	No rule	Rule 9: every named part enters the physics

Fault seen	Tests	Cause in v1	Correction in v2
Lamp drawn as a symbol when a realistic bulb was expected	T1 (first run)	Not specified	Rule 10: IEC 60617 symbols by default, realistic picture only when asked
"Upload a picture" read as an image-generation request	T6	Nothing said that every request is for a scene	Section 0: every request is an AnimLab scene, even when it mentions a picture

The v2 instructions are 30,145 characters (v1: 22,524). All four worked examples in v2 run without errors in AnimLab; example 5.4 gives H = 2.54 m and D = 10.19 m against a theory of 2.55 m and 10.19 m. The AI instructions page now serves v2.

Re-test with the corrected instructions (v2)

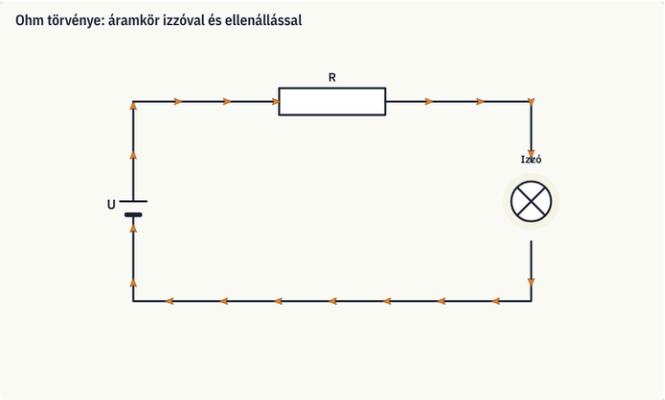
With v2, Gemini 3.5 Flash-Lite passed every re-tested prompt on the numbers; the remaining faults are small drawing details. The re-test ran the same evening in a new chat, same model, with the v2 text checked by checksum.

Test	Gemini v2 result	Reference	v1 → v2
T1 Ohm's law (U = 12 V, R = 6 Ω)	I = 0.57 A, P = 4.90 W (R_L = 15 Ω); arrows use `pos`, sit on wire pieces, vanish at 0 V; IEC symbols	I = 0.667 A, P = 5.33 W (R_L = 12 Ω)	Partial → Passed, with a 30 px gap above and below the lamp
T2 Transformer (230 V, 500/50, 10 Ω)	U2 = 23.0 V, I1 = 0.23 A, I2 = 2.30 A; `plotWindow(0.04)`; arrows use `pos`, hidden at zero	23.00 V, 0.230 A, 2.300 A	Partial → Passed, but the wire loops still run through the AC source and the resistor
T3 Projectile (10 m/s, 45°)	H = 2.54 m, D = 10.18 m, T = 1.44 s	H = 2.54 m, D = 10.18 m	Failed → Passed, but the code is almost a copy of worked example 5.4
T3b Bouncing ball (new, h0 = 5 m, e = 0.8)	h1 = 3.19 m against 3.20 m, 3 bounces in 4 s; ball starts at `y: h0 + r`, `trail: 500`	h1 = 3.18 m, 3 bounces	New test, Passed; the bounce count shows as "3,00" (no `digits` given)

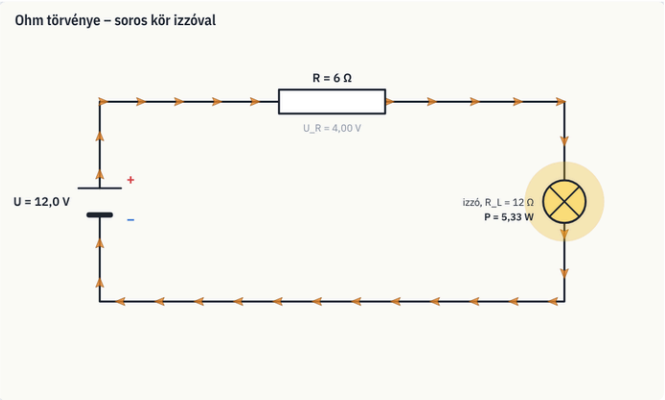
Test	Gemini v2 result	Reference	v1 → v2
T6 Rotor photo (first wording)	Code at once, asks for an upload named "rotor", ω = 6.28 rad/s	ω = 6.283 rad/s	Rewording needed → Passed; "É (N)" and "D (S)" pole labels are Hungarian only

T3 alone is weak evidence because its prompt matches the new example, so T3b was added: a physics-world task that no example covers. Gemini solved it with the right starting height, a numeric trail and bounce detection on ground contact, which are exactly the v2 rules.

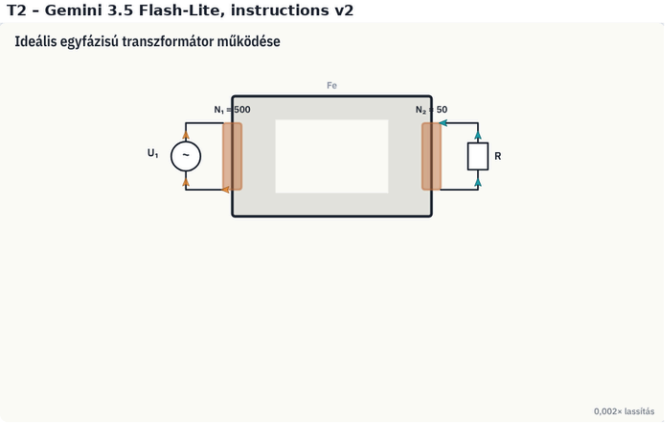
T1 - Gemini 3.5 Flash-Lite, instructions v2



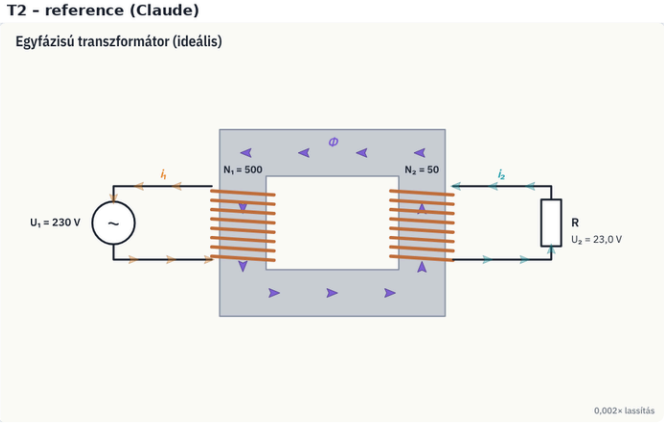
T1 - reference (Claude)



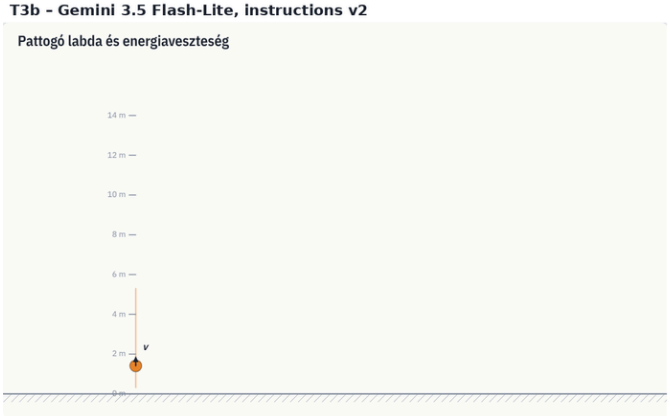
T2 - Gemini 3.5 Flash-Lite, instructions v2



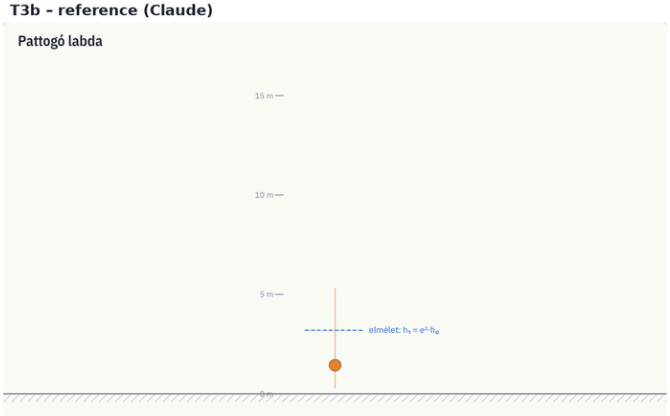
T2 - reference (Claude)



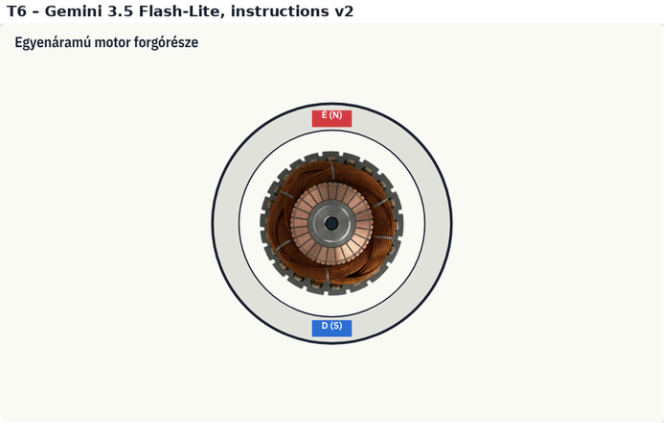
T3b - Gemini 3.5 Flash-Lite, instructions v2



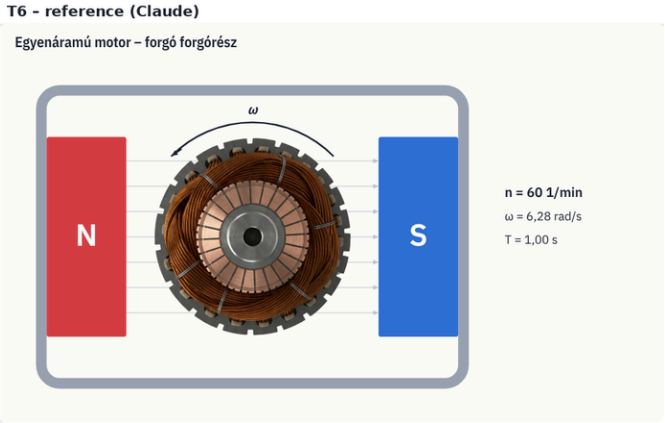
T3b - reference (Claude)



T6 - Gemini 3.5 Flash-Lite, instructions v2



T6 - reference (Claude)



What v2 does not fix yet: "stop at the component" is followed only roughly (gaps at the lamp, loops through the source and the resistor in T2), and a label inside a drawing was left Hungarian-only. A rule with a small drawing example (wire pieces meeting the lamp circle at y ± r) would be the next step. T4 and T5 were not repeated, since they passed with v1 and v2 does not touch them.

Files

Everything is in `Documents\AnimLab` on the Legion computer.

Folder	Contents
`app`	`animlab.html`, the current AnimLab with the rotor photo
`docs`	`animlab-ai-instructions.md` (v2, live on the AI instructions page) and `animlab-ai-instructions-v1.md`
`gemini-test-2026-11-03\gemini-session`	The v1 session transcript with every prompt, answer and code, and the instructions as sent
`gemini-test-2026-11-03\gemini-codes`	Gemini's codes T1–T6 for v1, plus the broken input used in T5
`gemini-test-2026-11-03\reference-codes`	Claude's reference codes T1–T6 and T3b
`gemini-test-2026-11-03\screenshots`	Every scene's screenshot and the side-by-side comparisons
`gemini-test-2026-11-03\v2`	The v2 re-test: transcript, Gemini codes, screenshots, results
`gemini-test-2026-11-03\harness`	The test scripts (`compare.js`, `compare-v2.js`, `lint.py`) and their raw results

Folder	Contents
`gemini-test-2026-11-03\test-report.pdf`	This report as a PDF